

M09 ENHANCED ACCESS TO NATURE | O (MAX : 2 PT)

Intent : Support access to nature beyond M02: Access to Nature and Beauty, by further incorporating nature through interior design, exterior design and access to nearby nature.

Summary : This WELL feature requires the integration of nature and natural elements into the interior and exterior of the project, as well as the provision of nature views and nearby nature, such as green and blue spaces.

Issue : Exposure to plants and other natural elements has been linked with decreased levels of diastolic blood pressure, depression and anxiety, increased attentional capacity, better recovery from job stress and illness, increased psychological well-being and increased pain tolerance.^{1,2} The incorporation of plants in the work environment is linked with improved employee morale, decreased absenteeism and increased worker efficiency and job satisfaction.^{1,3} The incorporation of water into built spaces can also relieve stress, promote satisfaction and enhance performance.⁴ Access to outdoor green spaces is associated with a range of short- and long-term mental health benefits for individuals across the life span, from children to older adults.⁵⁻⁷ The many associated benefits include lower levels of anxiety and depression, as well as improved mental recovery from stress and fatigue.⁸

Solutions : A dose-response relationship has been found with exposure to indoor nature, with research finding that as workday nature contact increased, perceived job stress, subjective health complaints and sickness absence decreased.⁹⁻¹¹ Researchers also hypothesize that access to outdoor green spaces impacts mental health and well-being through cognitive restoration, and that physiological and emotional changes take place, when individuals are exposed to natural settings.¹² The benefits of nature access have been found across diverse age groups, and supporting access to outdoor green and natural spaces can support the health and well-being of a wide range of individuals within a built community.^{13,14}

PART 1 PROVIDE NATURE ACCESS INDOORS (MAX : 1 PT)

For All Spaces:

The project's floor plan is designed such that at least 75% of workstations, conference room seats, classroom seats and seating within common spaces meets at least one of the following requirements:

- a. Have a direct line of sight to indoor plant(s), water feature(s) and/or nature view(s).
- b. Are within 10 m of indoor plant(s), water feature(s) and/or nature view(s).

Note :

Water features that may produce water aerosols (e.g., decorative fountains) cannot be installed in hospitals, healthcare settings or senior living facilities.¹⁵

PART 2 PROVIDE NATURE ACCESS OUTDOORS (MAX : 1 PT)

For All Spaces:

1: Outdoor nature

One of the following requirements is met:

- a. Outdoor nature access facilitated by the conditions below:
 1. One or more outdoor spaces that collectively total at least 5% of the project interior area are available to all regular occupants.
 2. At least 70% of the outdoor space must be covered by plants, water or other natural elements. Use the area of each tree or shrub canopy to calculate their contributions to the total area.
- b. Nearby nature access facilitated by the conditions below:
 1. At least one green space or blue space is within a 200 m walk distance from the project boundary and available to all regular occupants during open hours of the space(s).
 2. Total combined green space must be at least 0.5 hectare.¹⁶

2: Outdoor nature access

The following requirement is met:

- a. Occupants are encouraged to access outdoor nature (e.g., presence of signage or maps to outdoor nature, availability of breaks during the workday to go visit outdoor nature).

Note :

Interiors projects may count base building amenities towards feature requirements.

REFERENCES

1. Larsen L, Adams J, Deal B, Kweon B-S, Tyler E. Plants in the workplace the effects of plant density on productivity, attitudes, and perceptions. *Environ Behav.* 1998;30(3):261-281.
2. Wolf K, Krueger S, Flora K. Work and Learning - A Literature Review. *Green Cities Good Heal.* 2014. www.greenhealth.washington.edu. Accessed January 12, 2018.
3. Kant I, Beurskens a JHM, Amelsvoort LGPM Van, Swaen GMH. An epidemiological approach to study fatigue in the working population: the Maastricht Cohort Study. 2003;32-39.
4. Kellert SR, Calabrese EF. *The Practice of Biophilic Design.*; 2015. www.biophilic-design.com.
5. Pearson DG, Craig T. The great outdoors? Exploring the mental health benefits of natural environments. *Front Psychol.* 2014;5:1-4. doi:10.3389/fpsyg.2014.01178
6. Aggio D, Smith L, Fisher A, Hamer M. Mothers' perceived proximity to green space is associated with TV viewing

time in children: the Growing Up in Scotland study. *Prev Med (Baltim)*. 2015;70:46-49.

7. Wu YT, Prina AM, Jones A, Matthews FE, Brayne C. Older people, the natural environment and common mental disorders: Cross-sectional results from the Cognitive Function and Ageing Study. *BMJ Open*. 2015;5:1-9. doi:10.1136/bmjopen-2015-007936
8. Beyer KMM, Kaltenbach A, Szabo A, Bogar S, Javier Nieto F, Malecki KM. Exposure to Neighborhood Green Space and Mental Health: Evidence from the Survey of the Health of Wisconsin. *Int J Environ Res Public Health*. 2014;11(3):3453-3472. doi:10.3390/ijerph110303453
9. Bjornstad S, Patil GG, Raanaas RK. Nature contact and organizational support during office working hours: Benefits relating to stress reduction, subjective health complaints, and sick leave. *Work*. 2016. doi:10.3233/WOR-152211
10. Fjeld T, Veiersted B, Sandvik L, Riise G, Levy F. The Effect of Indoor Foliage Plants on Health and Discomfort Symptoms among Office Workers. *Indoor Built Environ*. 1998;7(4):204-209. <https://www.karger.com/DOI/10.1159/000024583>.
11. Largo-Wight E, Chen WW, Dodd V, Weiler R. Healthy Workplaces: The Effects of Nature Contact at Work on Employee Stress and Health. *Public Health Rep*. 2011;126:124-131. doi:10.2307/41639273
12. Staats H, Jahncke H, Herzog TR, Hartig T. Urban Options for Psychological Restoration: Common Strategies in Everyday Situations. *PLoS One*. 2016;11(1):1-24. doi:10.1371/journal.pone.0146213
13. Alcock I, White MP, Wheeler BW, Fleming LE, Depledge MH. Longitudinal Effects on Mental Health of Moving to Greener and Less Green Urban Areas. *Environ Sci Technol*. 2014;48(2):1247-1255. doi:10.1021/es403688w
14. White MP, Alcock I, Grellier J, et al. Spending at least 120 minutes a week in nature is associated with good health and wellbeing. *Sci Rep*. 2019;9(1):1-11. doi:10.1038/s41598-019-44097-3
15. Health Protection Surveillance Centre. National Guidelines for the Control of Legionellosis in Ireland - 8.4.1 Hospitals and Healthcare Institutions. Dublin, Ireland; 2009.
16. World Health Organization. Urban Green Spaces and Health: a Review of Evidence. *WHO Reg Off Eur*. 2016:80. http://www.euro.who.int/__data/assets/pdf_file/0005/321971/Urban-green-spaces-and-health-review-evidence.pdf?ua=1.